

Madhya Pradesh – Grade 10

2018 – Mathematics

Choose the correct option and write it in your answer book:

1. When $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ then the system of equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

Options:

- (a) Has two solutions
- (b) Has no solution
- (c) Has infinitely many solutions
- (d) Has unique solution

Answer: (c)

2. In equation $3x + 2y = 7$, if $y = 2$ then, write the value of x .

Options:

- (a) 1
- (b) 2
- (c) -1
- (d) 3

Answer: (a)

Solution:

$$3x + 2y = 7$$

$$3x = 7 - 2 \times 2$$

$$= \frac{3}{3} = 1$$

3. The discriminant of the quadratic equation $x^2 - 7x - 5 = 0$ is

Options:

- (a) 17
- (b) 69
- (c) 29
- (d) 20

Answer: (b)

Solution:

$$\Delta = b^2 - 4ac$$

$$a = 1 \quad b = -7 \quad c = -5$$

$$\Delta = (-7)^2 - 4 \times 1 \times -5$$

$$= 49 + 20$$

$$= 69$$

4. All squares are

Options:

- (a) Congruent
- (b) Equal
- (c) Similar
- (d) All of the above

Answer: (c)

Solution:

5. The sum of pairs of opposite angles of a cyclic quadrilateral is

Options:

- (a) 360°
- (b) 90°
- (c) 180°
- (d) 270°

Answer: (c)

Solution:

Write True/False in the following :

1. Algebraic expression $x^2 + 3\sqrt{x} - 4$ is not a polynomial.

Answer: True

2. Property tax is an indirect tax

Answer: False

3. The line joining the centre of a circle to the mid point of a chord is perpendicular to the chord.

Answer: True

4. Volume of cuboid = Length $\times$ Breadth $\times$ Height.

Answer: True

5. 2 is the mode of the following observations :

2,3,4,2,12,9,7,8,9,6,5,9.

Answer: False

Fill in the banks :

1. Third proportional of 8, 12 is _____

Answer: 18

Solution:

8: 12 :: 12 : x

$$\frac{8}{2} = \frac{12}{x}$$

$$x = \frac{144}{8}$$

$$= \frac{18}{12}$$

2. Compound amount = Principle $\times \left(1 + \frac{\text{Rate}}{100}\right)^{\text{Time}}$

Answer: Rate

3. Congruent figures are _____.

Answer: Similar

4. If two chords of a circle are equidistant from the centre of the circle then they are _____ to each other.

Answer: Equal

5. The probability of any sure event is always _____

Answer: 1

Write the answer in one word/ sentence of each:

1. Additive inverse of $x - \frac{1}{x}$ is

Answer: $-x + \frac{1}{x}$

2. $\log \frac{m}{n} =$

Answer: $\log m - \log n$

3. The edge of a cube is 5 cm. What will be its volume?

Answer: $vol = 5 \times 5 \times 5 = 125cm^3$

4. The relation between arc of the circle, angle subtended by the arc at the centre and radius is

Answer: Twice

5. Define Angle of Elevation.

Answer:

Angle of elevation is the angle subtended by the line of sight with the horizontal.

Match the correct pair :

Column (A)	Column (B)
(i) $\sin^2 45^\circ + \cos^2 45^\circ$	(a) 0
(ii) $\cot(90^\circ - \theta)$	(b) $\operatorname{cosec}^2 \theta$
(iii) $\cos \theta \cos(90^\circ - \theta) - \sin \theta \sin(90^\circ - \theta)$	(c) $\frac{\sqrt{3}}{2}$
(iv) $1 + \cot^2 \theta$	(d) 1
(v) $\cos 30^\circ$	(e) $\cos \theta$
	(f) $\tan \theta$
	(g) 2

Answer:

(i) – (d)

(ii) – (f)

(iii) – (a)

(iv) – (b)

(v) – (c)

6. What is the meaning of similarity?

OR

What are the conditions for the similarity of the polygons?

Solution:

(i) The two figures have same shapes.

(ii) There are two conditions for two polygons to be similar, all the corresponding angles must be equal and all the corresponding sides must be proportional.

7. The perimeters of two similar triangles are 30 cm and 20 cm respectively. If one side of one triangle is 12 cm, then find the length of corresponding side of other triangle.

OR

Write the statement of Basic Proportionality Theorem (Thales Theorem).

Solution:

(i) let $\triangle ABC \sim \triangle PQR$

$$\text{So, } \frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{PR} = \frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR}$$

$$\frac{30}{20} = \frac{12}{x}$$

$$x = \frac{20 \times 12}{30} = 8 \text{ cm}$$

(ii) If a line is parallel to a side of a triangle which intersects the other sides into two distinct points, then the line divides those sides in equal proportion.

8. Check whether 6 cm, 8 cm and 10 cm are the sides of right angled triangle?

OR

$\triangle ACB$ is an isosceles triangle such that $AC = BC$, if $AB^2 = 2AC^2$, then prove that $\triangle ACB$ is a right angled triangle.

Solution:

(i)

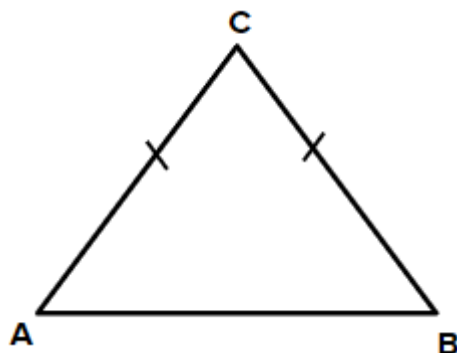
In right angled triangle Pythagoras holds true

$$6^2 + 8^2 = 36 + 64 = 100 \quad \text{LHS}$$

$$10^2 = 100 = 100 \quad \text{RHS}$$

$$\therefore \text{LHS} = \text{RHS}$$

(ii)



$$AC = BC$$

$$AB^2 = 2AC^2 \quad (\text{given})$$

To prove $\triangle ACB$ is right angled

$$AC^2 + BC^2 = AB^2$$

Proof : $LHS = AC^2 + AC^2 \quad (\because AC = BC)$
 $2AC^2$

$$RHS = AB^2 = 2AC^2 \quad (given)$$

Hence, it is true.

9. Find the median of the following values : 15, 35, 18, 26, 19, 25, 29, 20, 27.

OR

Write any two merits of Arithmetic mean.

Solution:

(i) Arrange data in ascending order

15, 18, 19, 20, 25, 26, 27, 29, 35

$$\frac{n+1}{2}th \quad (for \text{ odd})$$

$$\text{Median} = \frac{9+1}{2} = 5th$$

$$\text{so, median} = 25$$

(ii) (a) It is easy to calculate and simple to understand.

(b) It is based on all observations of given data

10. Find the probability that an even number turns up in a single throw of a die.

OR

Find the probability of getting head and tail at a time in single throw of a coin.

Solution:

(i) Sample space = {1, 2, 3, 4, 5, 6}

Even event = {2, 4, 6}

$$P(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

(ii) Sample space = {H, T}

Probability = 0

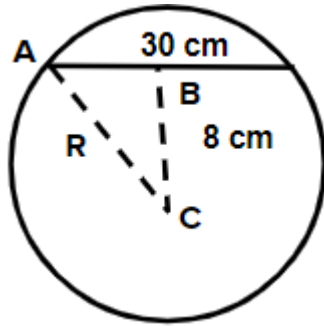
11. A chord of length 30 cm is drawn at a distance of 8 cm from the centre of the circle. Find the radius of a circle.

OR

Prove that, the length of two tangents drawn from an external point to a circle are equal.

Solution:

(i)



In $\triangle ABC$

$AB = 15\text{cm}$ (by theorem)

$\angle ABC = 90^\circ$

$$AB^2 + BC^2 = AC^2$$

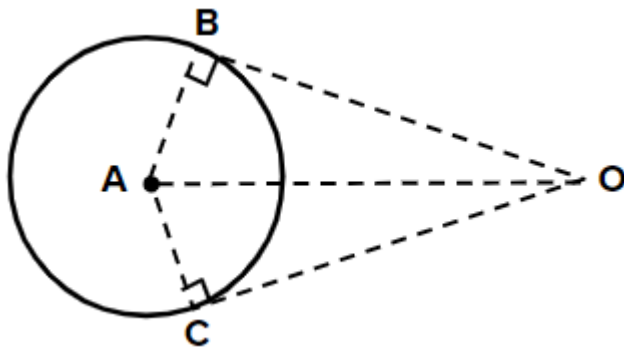
$$15^2 + 8^2 = AC^2$$

$$225 + 64 = AC^2$$

$$\sqrt{289} = AC$$

$$AC = 17\text{cm}$$

(ii)



In $\triangle ABO$ and $\triangle ACO$

$AO = AO$ (common)

$\angle ABO = \angle ACO$ (90°)

$AB = AC$ (Radii)

$\triangle ABO \cong \triangle ACO$ by RHS

So, $OB = OC$ by C.P.C.T

12. Prove that, the angles in the same segment of circle are equal to each other.

OR

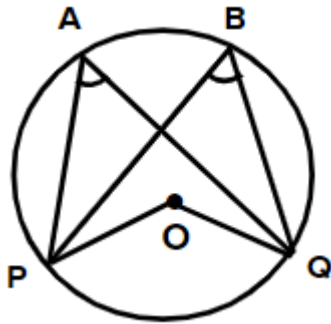
Prove that, the angle in the semicircle is right angle.

Solution:

(i) Given : A circle with center O. Points P and Q are on the circumference of this circle subtending angles $\angle PAQ$ and $\angle PBQ$ at points A and B respectively.

To prove : $\angle PAQ = \angle PBQ$

Proof:



Chord PQ subtends $\angle POQ$ at the centre from theorem 10.8 : Angle subtended by in one at the centre is double the angle subtended by it at any other point of circle.

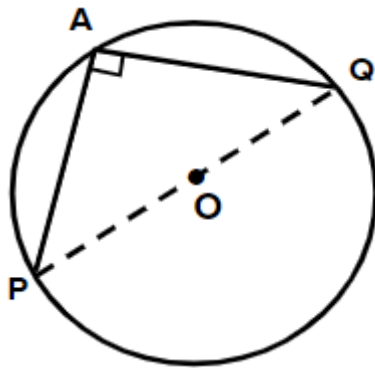
$$\therefore \angle POQ = 2\angle PAQ \quad (1)$$

$$\angle POQ = 2\angle PBQ \quad (2)$$

from (1) and (2)

$$\angle PBQ = \angle PAQ$$

(ii)



PQ is a diameter

Subtending $\angle PAQ$ at A on circle

To prove : $\angle PAQ = 90^\circ$

Proof: now,

POQ is a straight line passing through O

$\therefore$ Angle subtended by arc PQ at O is

$$\angle POQ = 180^\circ \quad (i)$$

By theorem: The angle subtended by an one at the centre is double the angle subtended by it at any point on remaining part.

Thus,

$$\angle POQ = 2\angle PAQ$$

$$\frac{\angle POQ}{2} = \angle PAQ$$

$$\angle PAQ = \frac{180}{2} = 90^\circ$$

13. If the mean of 5 observations $x, x + 2, x + 4, x + 6, x + 8$ is 11, then find the value of x .

OR

Find the arithmetic mean of 47,53,49,60,39,42,53,52,50,55

Solution:

(i)

$$\begin{aligned} \text{mean} &= \frac{\text{Total observations}}{\text{Numner of observation}} \\ &= \frac{x + x + 2 + x + 4 + x + 6 + x + 8}{5} \end{aligned}$$

$$55 = 5x + 20$$

$$5x = 35$$

$$x = 7$$

(ii)

$$\begin{aligned} \text{Arithmetic mean} &= \frac{\text{Total observations}}{\text{Numner of observation}} \\ &= \frac{47 + 53 + 49 + 60 + 39 + 42 + 53 + 52 + 50 + 55}{10} \\ &= \frac{500}{10} = 50 \end{aligned}$$

14. Find the median of the following frequency distribution table :

Variable x	4	6	8	10	12	14	16
Frequency f	2	4	5	3	2	1	4

OR

Write two merits and one demerit of median.

Solution:

(i)

X	F	Cumulative
4	2	2
6	4	6
8	5	11
10	3	14
12	2	16
14	1	17
16	4	21
	21	

$$\text{Median} = \left(\frac{n+1}{2} \right) \text{th} = \frac{21+1}{2} = 11\text{th}$$

So, median = 8

(ii)

Two merits of median

(a) median is defined as in case of mean

(b) it can be easily calculated and is easy to understand.

One demerit

(c) it is fluctuated much more by fluctuations of sampling than A.M

15. Solve the following system of equation by Substitution Method :

$$x + y = 7$$

$$3x - 2y = 11$$

OR

If in $\triangle ABC$, $\angle C = 2\angle B = \angle A + \angle B + 20$, then find all the three angles of the triangle.

Solution:

(a)

$$x + y = 7$$

$$y = 7 - x \quad (1)$$

$$3x - 2y = 11$$

$$3x - 2(7 - x) = 11$$

$$3x - 14 + 2x = 11$$

$$5x = 25$$

$$x = 5$$

$$y = 7 - 5 = 2$$

$$x = 5$$

$$y = 2$$

(ii) In triangle

$\triangle ABC$

$$\angle A + \angle B + \angle C = 180^\circ \quad (1)$$

$$\angle C = 2\angle B$$

$$2\angle B = \angle A + \angle B + 20$$

$$\angle B = \angle A + 20$$

$$\angle A + \angle A + 20 + 2\angle B = 180$$

$$\angle A + \angle A + 20 + 2\angle A + 40 = 180$$

$$4\angle A = 120$$

$$\angle A = 30$$

$$\angle B = \angle A + 20$$

$$= 30 + 20$$

$$= 50^\circ$$

$$\angle C = 2\angle B$$

$$= 2 \times 50 = 100^\circ$$

$$\angle A = 30^\circ, \angle B = 50^\circ, \angle C = 100^\circ$$

16. A fraction become $\frac{1}{4}$ when 2 is subtracted from its numerator and 3 is added to its denominator and on adding 6 to numerator and multiplying denominator by 3 it becomes $\frac{2}{3}$. Find the fraction.

OR

The sum of two numbers is 7. If the sum of these numbers is 7 times of its difference, then find the numbers.

Solution:

(i)

Let the fraction be $\frac{x}{y}$

$$\frac{x-2}{y+3} = \frac{1}{4}$$

$$4x-8 = y+3$$

$$4x-y = 11 \quad (1)$$

$$8x-2y = 22$$

$$\frac{x+6}{3y} = \frac{2}{3}$$

$$3x+18 = 6y$$

$$3x-6y = -18$$

$$x-2y = -6 \quad (2)$$

Solve equation (1) and (2)

$$8x-2y = 22$$

$$x-2y = -6$$

$$7x = 28$$

$$x = 4$$

$$y = 4x-11$$

$$= 4 \times 4 - 11$$

$$= 16 - 11$$

$$y = 5$$

(ii)

Let two number be x and y

$$x + y = 7 \quad (1) \quad x = 7 - y$$

$$x + y = 7(x - y) \quad (2)$$

$$7 - y + y = 7(7 - y - y)$$

$$7 = 7(7 - 2y)$$

$$1 = 7 - 2y$$

$$2y = 6$$

$$y = 3$$

$$x = 7 - 3$$

$$x = 4$$

17. If b is the mean proportional of a and c prove that $\frac{a^2 + b^2}{ab} = \frac{a + c}{b}$.

OR

If $\frac{x}{b+c} = \frac{y}{c+a} = \frac{z}{a+b}$ then prove that $(b-c)x + (c-a)y + (a-b)z = 0$.

Solution:

(i)

$$a : b :: b : c$$

$$b^2 = ac$$

$$\begin{aligned} LHS \quad \frac{a^2 + b^2}{ab} &= \frac{a^2 + ac}{ab} \\ &= \frac{a(a+c)}{ab} \\ &= \frac{a+c}{b} = RHS \end{aligned}$$

(ii)

$$\frac{x}{b+c} = \frac{y}{c+a} = \frac{z}{a+b} = k$$

$$x = k(b+c)$$

$$y = k(c+a)$$

$$z = k(a+b)$$

$$LHS = (b-c)x + (c-a)y + (a-b)z$$

$$(b-c)k(b+c) + (c-a)k(c+a) + (a-b)k(a+b)$$

$$k\{b^2 - c^2 + c^2 - a^2 + a^2 - b^2\}$$

$$= 0 = RHS$$

18. Solve the following equation formula method.

$$x^2 - 5x - 6 = 0.$$

OR

If α, β are the roots of quadratic equation $ax^2 + bx + c = 0$, then find the value of $\alpha^2 + \beta^2$.

Solution:

(i)

$$x^2 - 5x - 6 = 0$$

$$a = 1 \quad b = -5 \quad c = -6$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{5 \pm \sqrt{25 - 4 \times 1 \times -6}}{2 \times 1}$$

$$= \frac{5 \pm \sqrt{25 + 24}}{2} = \frac{5 \pm 7}{2}$$

$$= \frac{12}{2}, \frac{-2}{2} = 6 \text{ or } -1$$

(ii)

$$ax^2 + bx + c = 0$$

$$\alpha + \beta = \frac{-b}{a} \quad \alpha\beta = \frac{c}{a}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(\frac{-b}{a}\right)^2 - 2\frac{c}{a}$$

$$= \frac{b^2}{a^2} - \frac{2c}{a}$$

$$= \frac{b^2 - 2ac}{a^2}$$

19. At a point 25 m away from the foot of a building, the angle of elevation of the top of the building is 45° . Find the height of the building.

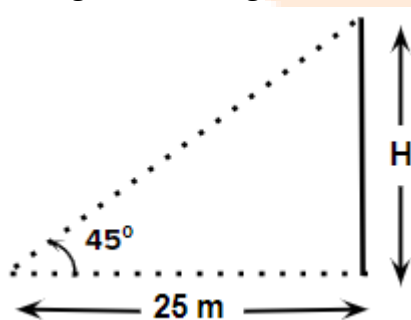
OR

From the top of 60 m high light house, the angle of depression of the ship is 60° . Find the distance between the ship and foot of the light house.

Solution:

(i)

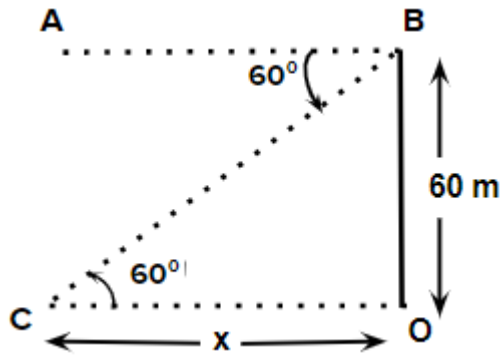
Let height of building be H



$$\tan 45 = \frac{H}{25}$$

$$H = 25m$$

(ii) let distance between light house and ship be x



$$\angle ABC = \angle BCO = 60^\circ$$

$$\tan 60^\circ = \frac{60}{x}$$

$$x = \frac{60}{\sqrt{3}}m$$

20. Find the length of arc and area of sector of the circle, if the angle subtended at the centre and the radius are 60° and 6 cm respectively.

OR

If V is the volume of cuboid whose length is a , breadth is b and height is c and S is its total surface area, then prove that $\frac{1}{V} = \frac{2}{S} \left[\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right]$.

Solution:

(i)

$$\begin{aligned} \text{Length of arc} &= \frac{\theta}{360} \times 2\pi r \\ &= \frac{60}{360} \times 2 \times \frac{22}{7} \times 6 \\ &= \frac{44}{7} \\ &= 6.28cm \end{aligned}$$

$$\begin{aligned} \text{Area of sector} &= \frac{\theta}{360} \times \pi r^2 \\ &= \frac{60}{360} \times \frac{22}{7} \times 6 \times 6 \\ &= \frac{132}{7} = 18.8cm^2 \end{aligned}$$

(ii)

Length = a

Breadth = b

Height = c (given)

Volume = l × b × h

V = a b c

Surface area = 2(lb + bh + hl)

S = 2 (ab + bc + ca)

$$\begin{aligned} RHS &= \frac{2}{S} \left[\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right] \\ &= \frac{2}{2(ab + bc + ca)} \left[\frac{ab + bc + ca}{abc} \right] \\ &= \frac{1}{abc} = \frac{1}{V} = LHS \end{aligned}$$

21. Two cubes each of 15 cm sides are joined end to end. Find the total surface area of the resulting cuboid.

OR

A iron sphere of radius 8 cm is melted then recasted into small spheres each of radius 1 cm. Find the number of small spheres.

Solution:

(i)

Length = 30 cm

Breadth = 15 cm

Height = 15 cm

$$\begin{aligned} \text{Surface area} &= 2(lb + bh + hl) \\ &= 2(30 \times 15 + 15 \times 15 + 15 \times 30) \\ &= 2(450 + 225 + 450) \\ &= 2250 \text{ cm}^2 \end{aligned}$$

(ii)

$$\text{Number of sphere} = \frac{\frac{4}{3} \pi \times 8^3}{\frac{4}{3} \pi \times 1^3} = 512$$

22. Which rational expression should be added to $\frac{x^3 - 1}{x^2 + 2}$ to get $\frac{2x^3 - x^3 + 3}{x^2 + 2}$?

OR

Factorise $a^2(b + c) + b^2(c + a) + c^2(a + b) + 3abc$.

Solution:

(i)

Let z be added

$$\begin{aligned}\frac{x^3-1}{x^2+2} + z &= \frac{2x^3-x^2+3}{x^2+2} \\ z &= \frac{2x^3-x^2+3}{x^2+2} - \frac{x^3-1}{x^2+2} \\ &= \frac{2x^3-x^2+3-x^3+1}{x^2+2} \\ &= \frac{x^3-x^2+4}{x^2+2}\end{aligned}$$

(ii) Factorise $a^2(b+c)+b^2(c+a)+c^2(a+b)+3abc$

{open all the parenthesis and distribute abc }

$$\begin{aligned}&= a^2b + a^2c + b^2c + ab^2 + c^2a + c^2b + 3abc \\ &= a^2b + a^2c + abc + b^2c + b^2a + abc + c^2a + c^2b + abc \\ &= a(ab+ac+bc) + b(bc+ab+ac) + c(ac+bc+ab) \\ &= (ab+ac+bc)(a+b+c)\end{aligned}$$

23. The sum of a number and its reciprocal is $\frac{26}{5}$. Find the number.

OR

Solve: $\sqrt{3x^2-2}+1=2x$.

Solution:

(i)

$$x + \frac{1}{x} = \frac{26}{5}$$

$$5(x^2+1) = 26x$$

$$5x^2 - 26x + 5 = 0$$

$$5x(x-5) - 1(x-5)$$

$$(x-5)(5x-1)$$

$$x = 5 \text{ or } x = 1/5$$

(ii)

$$\sqrt{3x^2-2}+1=2x$$

$$\left(\sqrt{3x^2-2}\right)^2 = (2x-1)^2$$

$$3x^2-2 = 4x^2+1-4x$$

$$0 = x^2 - 4x + 3$$

$$= x^2 - 3x - x + 3$$

$$= x(x-3) - 1(x-3)$$

$$(x-3)(x-1)$$

$$x = 3 \text{ or } x = 1$$

24. Find the depreciation and depreciated cost of a motor cycle after 3 years at the rate 10% depreciation which costs Rs. 40,000.

OR

Find the amount and compound interest on Rs.2,000 for 2 years at the rate of 10% per annum compound interest.

Solution:

(i)

$$P = 40000$$

$$R = 10\%$$

$$T = 3 \text{ years}$$

$$D.C = P \left(1 - \frac{r}{100} \right)^n$$

$$= 40000 \left(1 - \frac{10}{100} \right)^3$$

$$= 40000 \times \frac{90}{100} \times \frac{90}{100} \times \frac{90}{100}$$

$$= 29160$$

$$\begin{aligned} \text{Depreciation} &= 40000 - 29160 \\ &= 10840 \text{ Rs.} \end{aligned}$$

(ii)

$$P = 2000$$

$$R = 10$$

$$T = 2 \text{ years}$$

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$$= 2000 \left(1 + \frac{10}{100} \right)^2$$

$$= 2000 \times \frac{110}{100} \times \frac{110}{100}$$

$$= 2420 \text{ Rs}$$

$$CI = 2420 - 2000 = 420 \text{ Rs.}$$

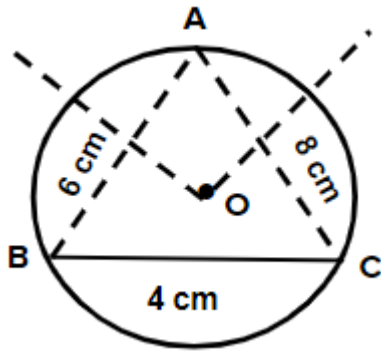
25. Construct a triangle whose sides are 4 cm, 6 cm and 8 cm. Draw the circumcircle of the triangle. Write the steps of construction also.

OR

Construct a triangle ABC, in which $BC = 6.5 \text{ cm}$, $\angle A = 60^\circ$ and median $AD = 4.5 \text{ cm}$.

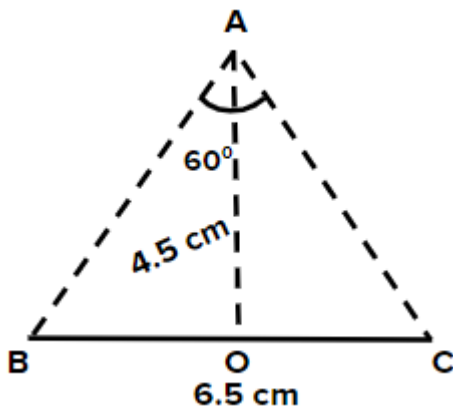
Solution:

(i)



- (i) Draw base $BC = 4$ cm
- (ii) Draw arc of 6 cm and 8 cm using compass
- (iii) Join them to get triangle
- (iv) Draw $\perp$ bisector of all sides
- (v) Where these intersect name it as O.
- (vi) Pointing on this draw circle

(ii)



Draw a line of 6.5 cm

Draw angle of 60°

Take mid points on 6.5 cm and Draw $4.5 = AO$

26. Prove it without using table $\frac{\sin 70^\circ}{\cos 20^\circ} + \frac{\cos ec 20^\circ}{\cos 70^\circ} - 3 \cos 70^\circ \cos ec 20^\circ = 0$.

OR

Prove that $\frac{1 + \cos A}{\sin A} + \frac{\sin A}{1 + \cos A} = \frac{2}{\sin A}$.

Solution:

(i)

$$\begin{aligned} & \frac{\sin 70}{\cos 20} + \frac{\cos ec 20}{\sec 70} - 2 \cos 70 \cos ec 20 \\ & \frac{\cos(90-70)}{\cos 20} + \frac{\sec(90-20)}{\sec 70} - 2 \cos 70 \times \sec(90-20) \\ & \frac{\cos 20}{\cos 20} + \frac{\sec 70}{\sec 70} - 2 \times \cos 70 \times \sec 70 \\ & 1 + 1 - 2 \times \cos 70 \times \frac{1}{\cos 70} \\ & 2 - 2 = 0 = RHS \end{aligned}$$

(ii)

$$\begin{aligned} & \frac{1 + \cos A}{\sin A} + \frac{\sin A}{1 + \cos A} \\ & = \frac{(1 + \cos A)^2 + \sin^2 A}{\sin A(1 + \cos A)} \\ & = \frac{1 + \cos^2 A + 2 \cos A + \sin^2 A}{\sin A(1 + \cos A)} \\ & = \frac{1 + 2 \cos A + 1}{\sin A(1 + \cos A)} = \frac{2(1 + \cos A)}{\sin A(1 + \cos A)} = \frac{2}{\sin A} = RHS \end{aligned}$$